

ACHIEVE!



SCIENCE

Living Things

Ally Chumley



Made for the
Australian Curriculum

Achieve! Science

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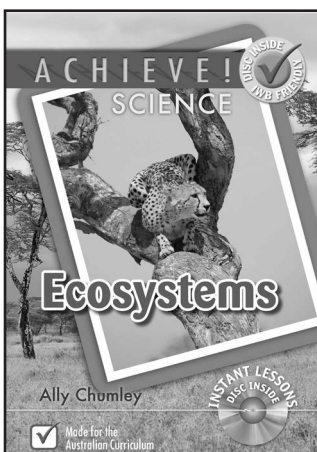
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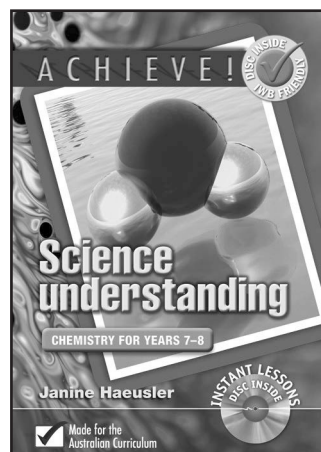
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Indigenous contexts

ISBN 978 1 92168 064 9



A C H I E V E !

SCIENCE

Living Things

Ally Chumley

This book was commissioned by Jennifer Gough
Project management and editing by Render Editorial Services
Typesetting by digiXL
Cover design by Chilli Moon Design

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Ally Chumley writes educational resources for teachers and learners across a range of subjects. She is an experienced teacher and enjoys the challenge of making difficult concepts accessible and inspiring in the classroom. Her formal credentials include a Bachelor degree and a Master of Education. Ally has written more than 40 titles for Australia's leading educational publishers. She lives in Comboyne, New South Wales, with her husband, two children and assorted farm animals.

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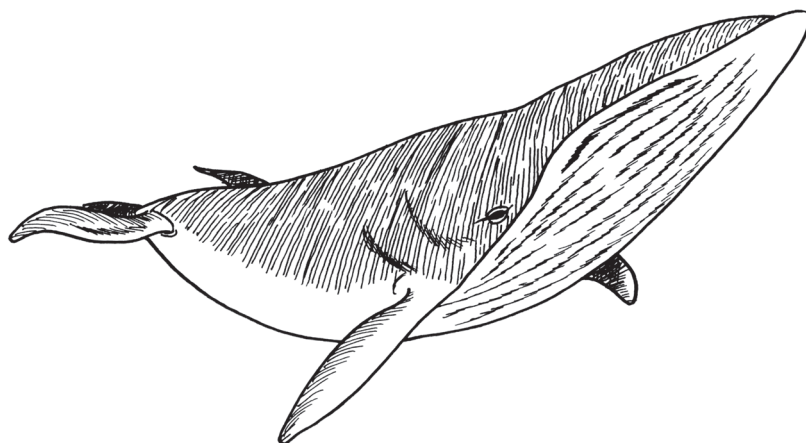
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ISBN 978 1 92571 063 2

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The *Achieve!* series of science titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students who struggle to read and comprehend regular classroom materials because of poor literacy skills. Many of the pages will also be suitable as supplementary material for regular classes.

Living things looks at the structure of animal and plant cells, how cells work, organs, important body systems, photosynthesis and the many ways to classify living things.

This book is divided into six units, all of which contain photocopiable Activity sheets and a Revision sheet. The Activity sheets can be used sequentially or as standalone work sheets. Varied activities – including labelling diagrams, completing tables, multiple-choice questions, cloze activities and word games – allow students to consolidate their understanding and improve literacy skills. A background information box defines key terms and provides a pronunciation guide. Each unit concludes with a Revision sheet to test understanding of the unit content. Answers have been included for easy and efficient marking.

The CD-ROM contains all the pages that are featured in the book. These pages can be easily edited to suit the individual needs of students. The CD-ROM also contains a folder of illustrations that can be used on other work sheets.

Australian Curriculum alignment		Unit					
	Year level	1	2	3	4	5	6
There are differences within and between groups of organisms; classification helps organise this diversity (ACSSU111)	7			✓			
Cells are the basic units of living things and have specialised structures and functions (ACSSU149)	8	✓		✓	✓		
Multi-cellular organisms contain systems of organs that carry out specialised functions that enable them to survive and reproduce (ACSSU150)	8		✓		✓	✓	
Multi-cellular organisms rely on coordinated and interdependent internal systems to respond to changes to their environment (ACSSU175)	9		✓			✓	✓
Chemical reactions, including combustion and the reactions of acids, are important in both non-living and living systems and involve energy transfer (ACSSU179)	9				✓		



What are cells?

All living things are made up of tiny parcels of matter called **cells**. Cells are the smallest units of living things. This is why they are sometimes called the **building blocks** of life.

Living things may be made up of just one cell or many billions of cells. Anything that is made up of one or more cells is considered a form of life or **lifeform**.

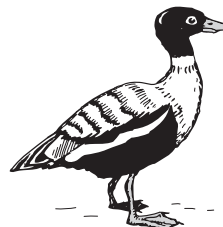
The bodies of all types of animals, including humans, are made up of cells. Different types of cells have different **functions** or jobs. Human bodies are made up of hundreds of different types of cells, including types of blood cells, nerve cells and skin cells.

All types of plants are made up of cells, too. Plant cells and animal cells have some similar parts and some different parts.

 **1** Which of these things are living things? Circle your choices.



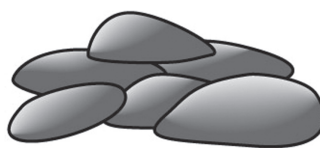
a) cake



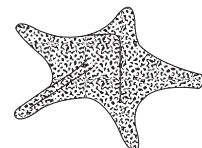
b) duck



c) fern



d) pile of stones



e) sea star

2 Which of the things in Question 1 are made up of cells?

3 Circle the correct phrase within the brackets to complete each sentence.

- a)** Cells are (tiny parcels of matter / the enemies of all living things / found only in human bodies). They are sometimes called the (playdough / building blocks / organisms) of life.
- b)** Different types of cells have (different coloured parts / the same purpose / different functions or jobs).

What are cells? CONT'D

4 Name three or more types of cells found in the human body.

When scientists talk about a living thing, they use the word **organism**. When they talk about things related to cells, they use the word **cellular**.

5 Write the words in the box into the answer spaces to break them into syllables. Remember, each syllable makes one beat when you say the word aloud. For example, the word *matter* has two syllables: m a t / t e r.

living organism cellular units

- a) ____ / ____
- b) ____ / ____ / ____ / ____
- c) ____ / ____ / ____
- d) ____ / ____

6 If 'multi-' means many and 'uni-' means one, complete these sentences using the words from the word bank.

- a) A _____ thing made up of _____
_____ is called a multicellular organism.
- b) A living thing made up of _____ is
called a unicellular _____.

Word bank

organism
cells
one
living
cell
many

The structure of cells

BACKGROUND

nucleus (say 'nu-kle-us') – the round structure at the centre of the cell containing the genetic information that controls what the cell does

cytoplasm (say 'site-o-plaz-em') – the main fluid substance inside a cell, which helps protect the cell's contents

cell membrane (say 'sell mem-brain') – the thin skin around an animal or plant cell that holds the cell together

A cell is like a tiny parcel that contains many different parts or **structures**. All types of animal and plant cells contain the following three structures:

- the nucleus
- the cytoplasm
- the cell membrane.



1 Add the missing consonants to these letters to make the names of cell structures. Use the meanings to help you.

Letters	Name	Meaning
u e u		found in the centre of the cell
e e a e		a thin skin
y o a		the jelly-like stuff inside a cell



2 Complete this passage by filling in the missing double letters. Hint: It is the same set of double letters that is missing throughout.

Both animal and plant ce ____ s are held together by a soft ce ____ membrane. However, plant ce ____ s have another outer layer, called the ce ____ wa ____, which is made of a substance called ce ____ ulose. These hard ce ____ wa ____ s give plants their shape and make them able to stand up ta ____ . Animal ce ____ s do not have ce ____ wa ____ s.

The structure of cells CONT'D

3 Choose the correct word within the brackets to complete each sentence.

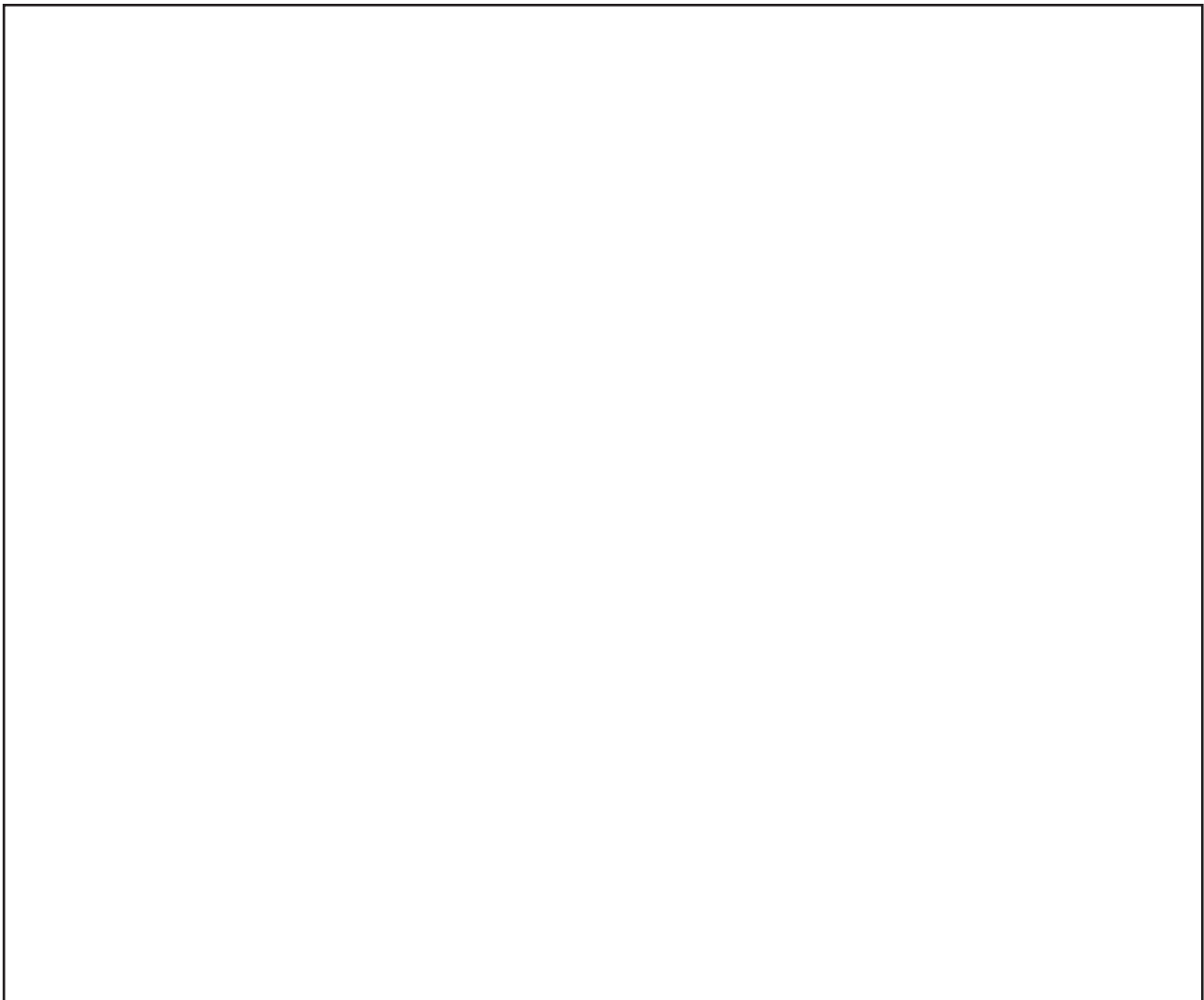
- a)** A leaf is made up of tiny units of matter called _____. (cells / cellulose)
- b)** Many cells make up the _____ of human beings. (nucleus / bodies)
- c)** The cells of humans and animals have _____ cell walls. (hard / soft)

4 Circle the correct answer.

What are the walls of plant cells made of?

cytoplasm cellulose cells

5 Using the information in this work sheet, draw a basic diagram of a plant cell. Label the four parts described in this Activity sheet.



Animal cells

Most types of animal cells have a centre called a nucleus near their middle, which is surrounded by a jelly-like substance called cytoplasm. Around this is the cell membrane.

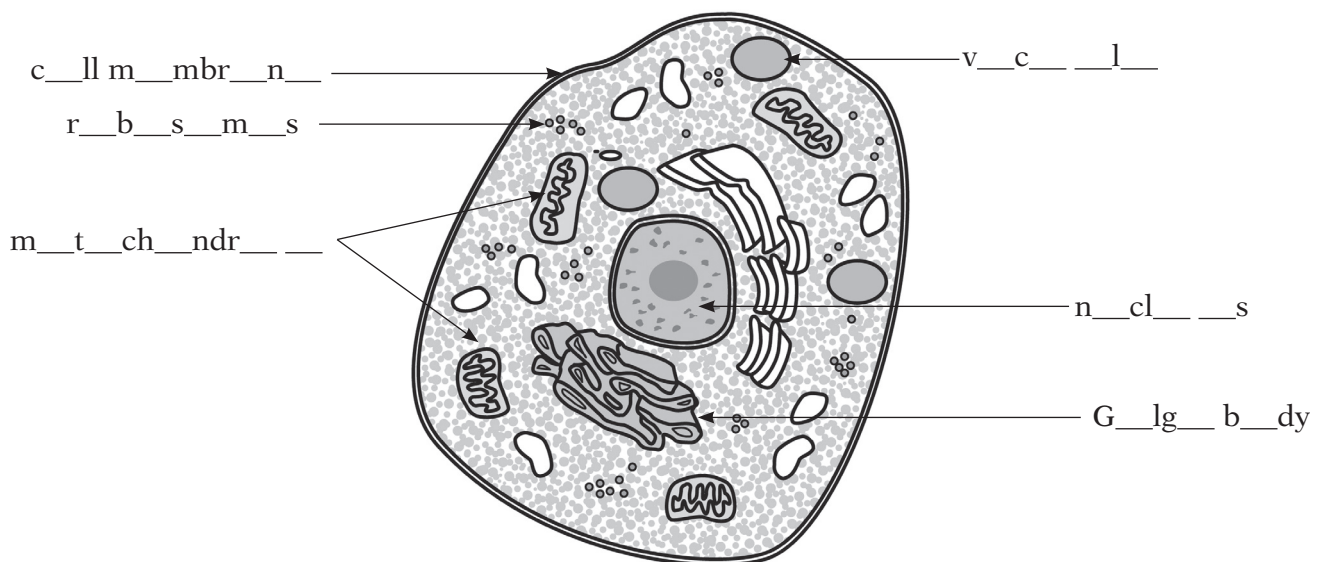
Cells also contain things called **proteins** and **organelles**. Proteins are a special kind of nutrient that are an essential part of all living things. Organelles are tiny, living machines that are found inside cells. The organelles perform various jobs to keep the cell alive. Types of organelles include:

- mitochondria, which provide energy for the cell to use
- ribosomes, which are like protein factories
- vacuoles, which are water-filled organelles that can have many functions
- Golgi bodies, which process proteins.

 **1** Look at the diagram of the animal cell.

a) Fill in the missing vowels (a, e, i, o, u) to complete the labels.

b) Add a label and pointer for cytoplasm.



2 Complete the names of three types of animal cells.

- | | |
|--------------------|---|
| b __ __ __ cells | flow through veins and arteries |
| n __ __ v __ cells | pass electrical messages around the body to help the body feel things |
| __ k __ __ cells | make up the outer covering of a body |

Plant cells

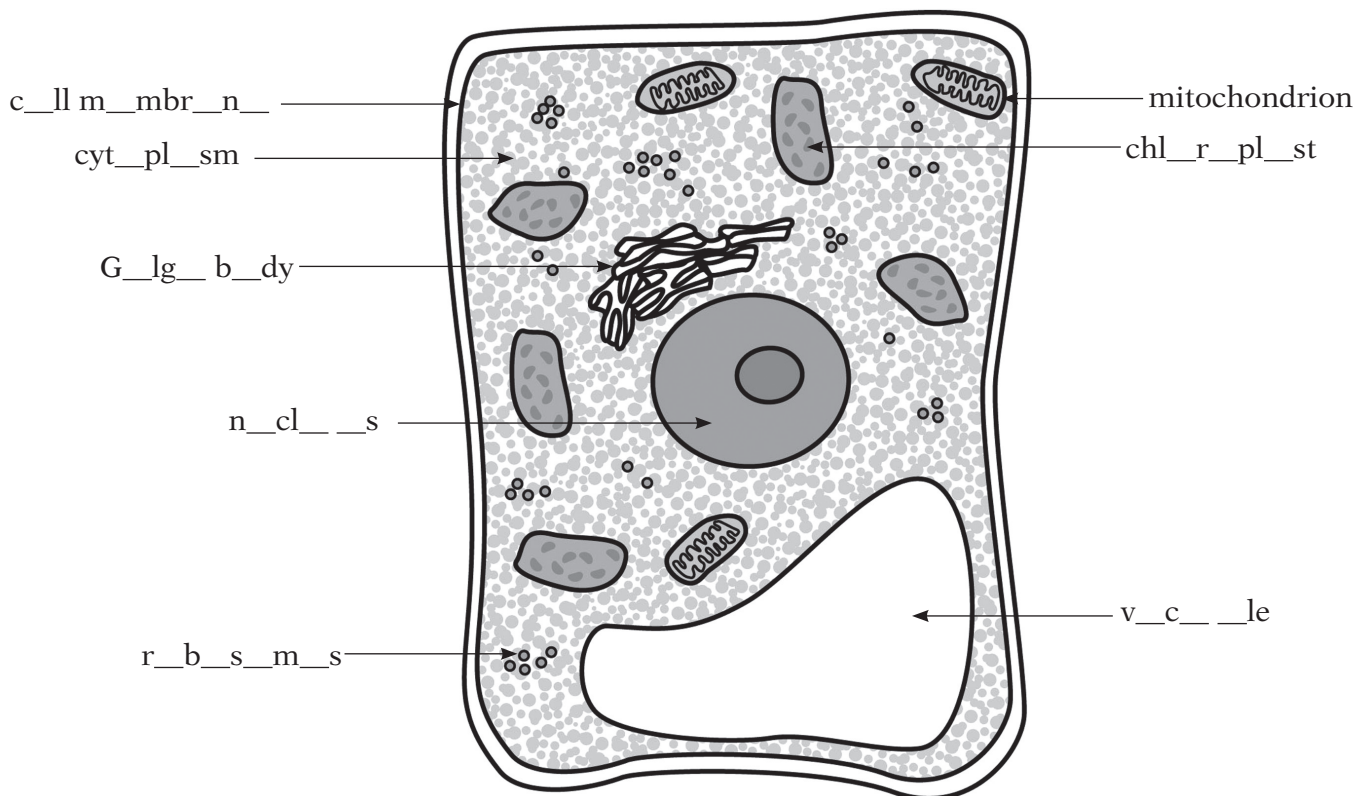
Like animal cells, most plant cells have a nucleus, cytoplasm and cell membrane. Plant cells also have a hard cell wall made of cellulose. Also like animal cells, plant cells contain tiny, living machines called organelles. Golgi bodies and ribosomes are two organelles found in plant cells. They have special jobs to do in the cell to help the plant grow and to keep it healthy. Plant cells also contain large water-filled containers called vacuoles.

Green plants contain important organelles called **chloroplasts**. These parts have a chemical called **chlorophyll** in them. It is green. Green plants get their green colour from the chloroplasts. Chloroplasts are involved in the way that green plants use sunlight to produce food, through photosynthesis.

 **1** Look at the diagram of the plant cell below.

a) Fill in the missing vowels to complete the labels.

b) Add a label and pointer for the cell wall.



2 Complete the following words.

chloro _ _ _ _ _

A green chemical pigment found in green plant cells

chloro _ _ _ _ _

An organelle found in green plant cells

What do cells do?

 **1** Fill in the blanks with the correct word from the word bank.

	Word bank
Cells hold all of the instructions and equipment that are needed to maintain life. Different types of cells perform different _____, but most living things need billions of _____ cells in order to function and _____. For example, scientists estimate that there are about 37.2 trillion cells in the adult human body.	healthy grow jobs
The instructions in a cell are held in the _____ (in genes) and the organelles do the work for the cell, such as turning _____ into the materials that a body or plant needs to make _____.	nutrients energy nucleus
In green _____ cells, the chloroplasts turn sunlight, carbon dioxide (a gas), _____ and nutrients into food for the cell to use.	water plant
In _____, many types of chemical processes happen in cells. Cells use the nutrients that animals get from _____ to perform different tasks needed by the _____. They release waste products. The _____ products are collected from the cells by the bloodstream and are removed from the body.	body food waste animals

2 Circle all the words that could correctly finish these sentences.

a) Cells keep a plant: healthy human growing dying living

b) Plant cells make
food from: sunlight animals nucleus water nutrients

Cell division

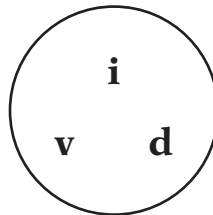
Cells do not last forever. They grow old and eventually die, so they need to be replaced. For example, skin cells in humans get worn out and are replaced continually. The flaky bits of dry skin that you scrub away are made of dead skin cells. Underneath this skin, healthy new skin cells are ready to replace the lost cells.

Cells also make living things grow bigger or stronger, by producing more of themselves. For example, a plant grows taller and gets more leaves as new cells are made.

To make more of themselves, cells separate into two parts that are the same. This is called **cell division**. Its scientific name is **mitosis**.

 **1** Use the letters in the circle to complete these forms of the word *division*.

- a) _ _ _ ide
- b) _ _ _ ides
- c) _ _ _ iding
- d) _ _ _ ided
- e) _ _ _ ision



2 Circle each of the correct phrases within the brackets to complete this explanation of mitosis.

Cells in living things make more of (themselves / proteins) in order to (renew / replace) old cells and (kill / create) new cells for growth. This process is called mitosis. Through mitosis, a (cell / plant) splits into (three / two), creating an (identical / adult) copy of itself.

3 Add the letters 'c' or 'e' to complete the words in this sentence, then rewrite the sentence in the space below.

Oth__r nam__s for mitosis ar__ _ _ _ll division and _ _ _ll r__produ__tion.

Check your understanding

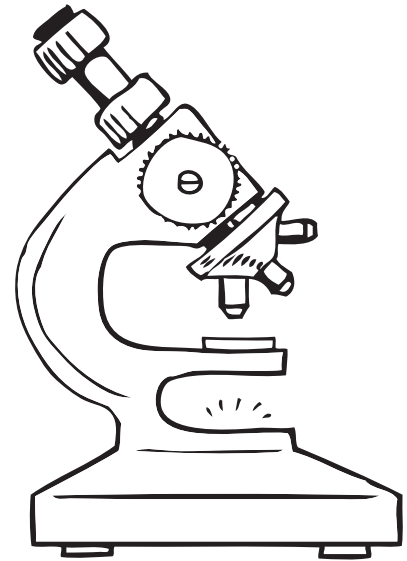
Circle the correct answers to these revision questions.

- 1** All living things are made up of tiny parcels of matter, which we call
 - a)** cytoplasm.
 - b)** cells.
 - c)** organisms.
- 2** Which sentence is correct?
 - a)** Plant cells have a soft cell membrane but also a hard cell wall.
 - b)** Plant cells have a soft cell wall, called the cell membrane.
 - c)** Animal cells have a hard cell wall, called the cell membrane.
- 3** What do most animal cells have at their centre?
 - a)** a nucleus
 - b)** a cell wall
 - c)** chloroplasts
- 4** What is cytoplasm?
 - a)** an organelle that provides energy for the cell
 - b)** a jelly-like substance inside a cell
 - c)** the material that cell walls are made of
- 5** Which of these pairs are types of cell organelles?
 - a)** energy and protein
 - b)** mitochondria and ribosomes
 - c)** Golgi bodies and cells
- 6** Plants get their green colour from which substance?
 - a)** blood
 - b)** cytoplasm
 - c)** chlorophyll
- 7** Cells need to get rid of which of the following?
 - a)** cell membranes
 - b)** organelles
 - c)** waste products
- 8** What is mitosis?
 - a)** cell reproduction
 - b)** cell death
 - c)** cell disease

How do we know about cells?

Cells are tiny parcels of matter that can't be seen with the naked eye. Before people knew about tiny, invisible things such as cells, bacteria (or germs) and viruses, they believed that bad spirits caused diseases.

Over the years, different kinds of **microscopes** have been invented. In the late 1600s scientists began to see cells for the first time. Scientists today use powerful microscopes to find out more about cells, such as what different cells do and how they work. This has helped us to treat illnesses and show people how to stay healthy.



-  **1** Match each event with the year it occurred, by writing each year from the box beside an event.
Hint: Think about the order in which these events would have happened.

1595

1838–39

1663

- a)** Robert Hooke first discovers the cell. _____
- b)** Zacharias Janssen invents the compound microscope. _____
- c)** Theodor Schwann and Matthias Schleiden state that plants and animals are made up of cells. _____
- 2 a)** *Microscopic* is a synonym of *tiny*. What is the base word of *microscopic*? Hint: It is mentioned in this Activity sheet. _____
- b)** List any more 'micro-' words that you can think of.

How do we know about cells? CONT'D

3 Write synonyms for the following words. A synonym is a word that has the same meaning as another word.

a) tiny _____

b) diseases _____

c) invented _____

d) powerful _____

e) work _____

f) healthy _____

4 Circle the correctly spelled word in each set of words.

a) parscle parsel parcel

b) bacteria backeria bateria

c) dieisease disease diease

d) microscope microsope micoscrope

5 Using your new words from Question 3, rewrite the opening paragraphs from this Activity sheet.

What jobs do cells do in humans?

BACKGROUND

cell membrane (say 'sell mem-brain') – the thin skin around an animal or plant cell that holds the cell together

cytoplasm (say 'site-o-plaz-em') – the main fluid substance inside a cell, which helps protect the cell's contents

The cells in humans have all sorts of jobs. For example, red blood cells carry oxygen gas around our bodies to keep our other cells working properly. There are around 200 different types of cells in the human body.

The soft **cell membrane** acts like a bag full of little holes. The cell lets some substances pass through the membrane, with some moving into the cell and some moving out of the cell. Other things are kept out completely.

 **1** Fill in the blanks with the correct word from the word bank.

Word bank

membrane
oxygen
microscope
jobs
cells

a) The word _____ means a gas that is in the air we breathe.

b) The word _____ means tasks.

c) The word _____ means the thin skin that holds a cell together.

d) The word _____ means tiny parcels of matter that make up living things.

e) The word _____ means a device that helps us see cells and their contents.

2 What things might a cell let in through the cell membrane?

a) nutrients and oxygen

b) waste products and carbon dioxide

c) cytoplasm and bacteria

3 What might a cell move out through the cell membrane to keep the body healthy?

a) nutrients and oxygen

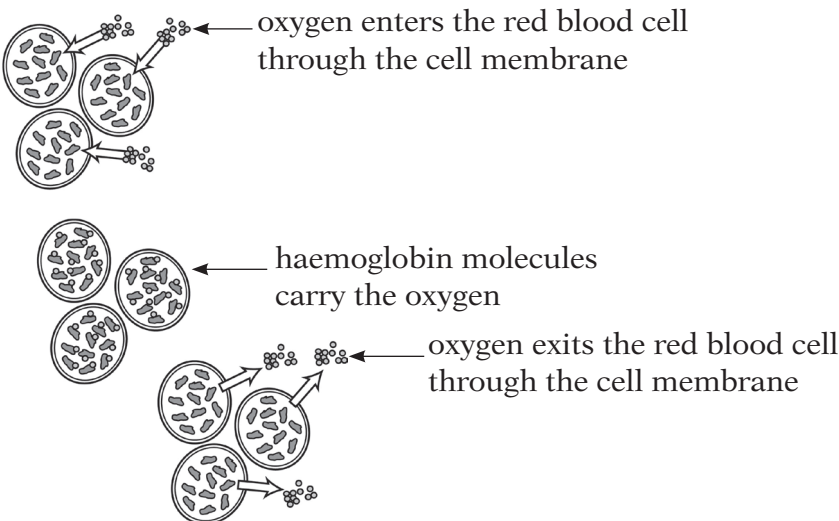
b) waste products and carbon dioxide

c) cytoplasm and bacteria

What jobs do cells do in humans? CONT'D

4 Rewrite this passage about the function of red blood cells, adding in all the spaces.

Red blood cells carry oxygen to the rest of the body's cells. The cell membrane lets in oxygen, and a substance called haemoglobin holds onto it tightly. When the red blood cell reaches other cells, it delivers the oxygen through the cell membrane.



Red blood cells transport oxygen around the body.

- 5 What important gas do cells need to survive? _____
- 6 In this table, nouns (names of things) are being transformed into adjectives (describing words). Fill in the blank spaces to complete the adjectives.

Noun	Adjective
cell	cellu _ _ r
body	bodi _ _
gas	gaseo _ s
microscope	microsc _ p _ c

Tissues, organs and body systems

Similar types of cells group together in an animal's body to form a **tissue**, such as a muscle, which carries out a particular job. Different tissues group together to form an **organ**.

An organ is a complete unit of the body that has a particular role or job to do. For example, the human heart is a fist-sized organ made up of cardiac muscle and connective tissues. Its function is to pump blood around the body.

Two or more organs work together in groups called **body systems**.

There are body systems for all sorts of things – including for breathing, for moving blood around the body and for digesting food. All the cells in a system need to do their jobs to make the whole system work.

-  **1** The three organs in this table are found in the human body. In the second column of the table, list anything you know about their function, or what they do.

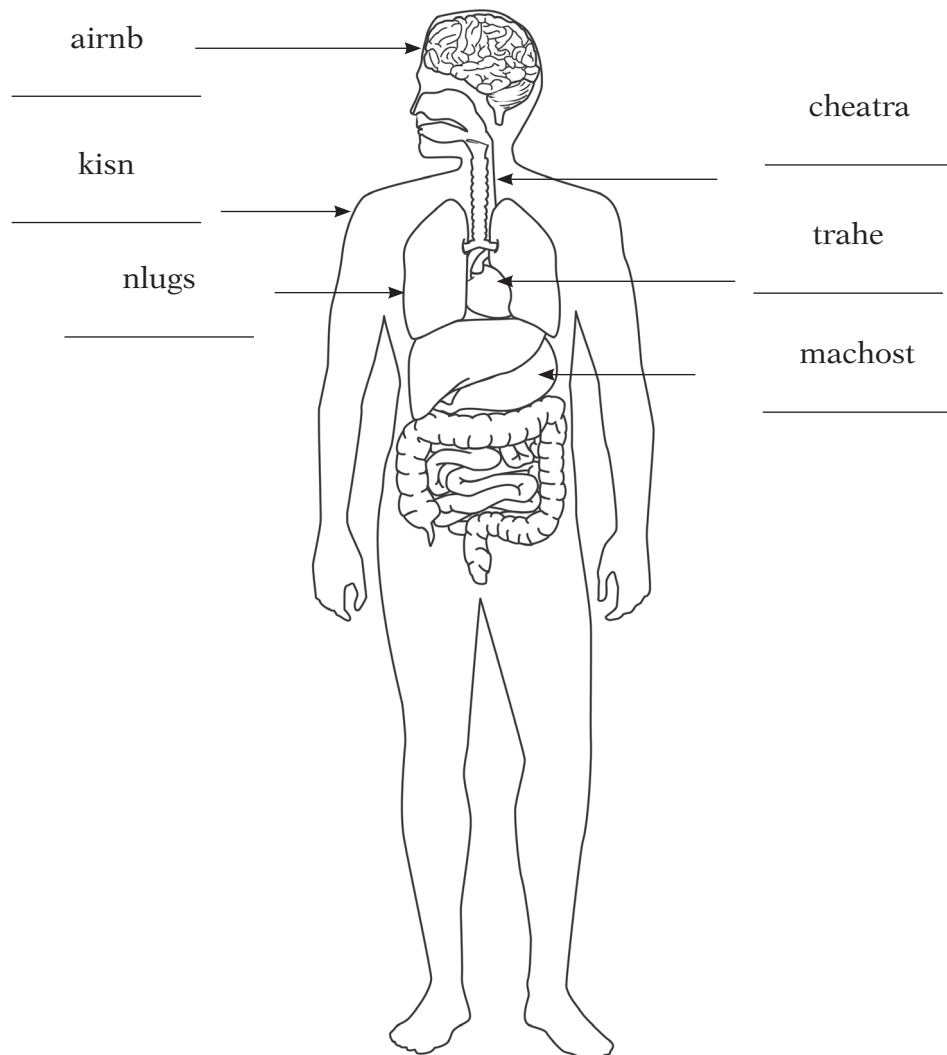
Organ	Function
heart	
lungs	
skin	

Tissues, organs and body systems CONT'D

2 Are systems made up of organs or are organs made up of systems? Write your answer in a full sentence.

3 List three other human organs. If you can, name the system each one belongs to.

4 Look at the diagram of the human body. Unscramble the words to identify some of the major organs.



Body systems

The bodies of animals and humans have billions of cells that are organised into tissues and organs. Groups of organs make up **body systems**. These systems help animals and humans stay healthy and active.

The body systems in an organism are **interdependent**. This means that each of the systems relies on the other systems. For example, the respiratory system supplies the human body with oxygen, and the circulatory system transports this oxygen around the body to the cells.

 **1** Draw a line between the name of each of these human body systems and its description.

- | | |
|----------------------------------|--|
| a) central nervous system | the system that removes waste products from the body |
| b) circulatory system | the system of bones, cartilage, ligaments and tendons |
| c) digestive system | the system that includes the sex organs |
| d) reproductive system | the system that passes electrical signals to and from the brain |
| e) respiratory system | the system that transports blood around the body |
| f) excretory system | the system that deals with the taking in and processing of food |
| g) skeletal system | the system that includes the organs used for breathing, or respiration |

2 a) In your own words, write a definition for *interdependence*.

b) Write down two body systems from Question 1 that you think are interdependent. Explain why.

The circulatory system

-  1 These passages are about a body system that is common to many animals: a circulatory system. Fill in each blank with the correct word from the word bank.

	Word bank
The design of an individual body system is different for _____ types of animals, but the function of some are the _____.	same different
Most animals – including _____ – have a circulatory system. A circulatory system transports nutrients, _____ and other gases, and waste products around a _____.	body humans oxygen
Simple animals, such as sea sponges, have _____ circulatory systems. Larger and more _____ animals have _____ and more complex circulatory systems.	larger complex simple
For vertebrates (animals with a backbone), such as humans, fish, birds and reptiles, the major _____ in the circulatory system is the heart. The heart pumps _____ around the body. The blood carries oxygen to all the _____ in the body, and it carries carbon dioxide away from the cells.	blood organ cells

- 2 In your own words, explain the statement *Body systems are interdependent*.

The central nervous system

Look at the diagram of the human central nervous system. The central nervous system passes electrical signals around the body, to and from the brain. It coordinates all the activities of the body, from breathing to running to winking.

-  **1** Finish these sentences by unscrambling the words in the brackets.

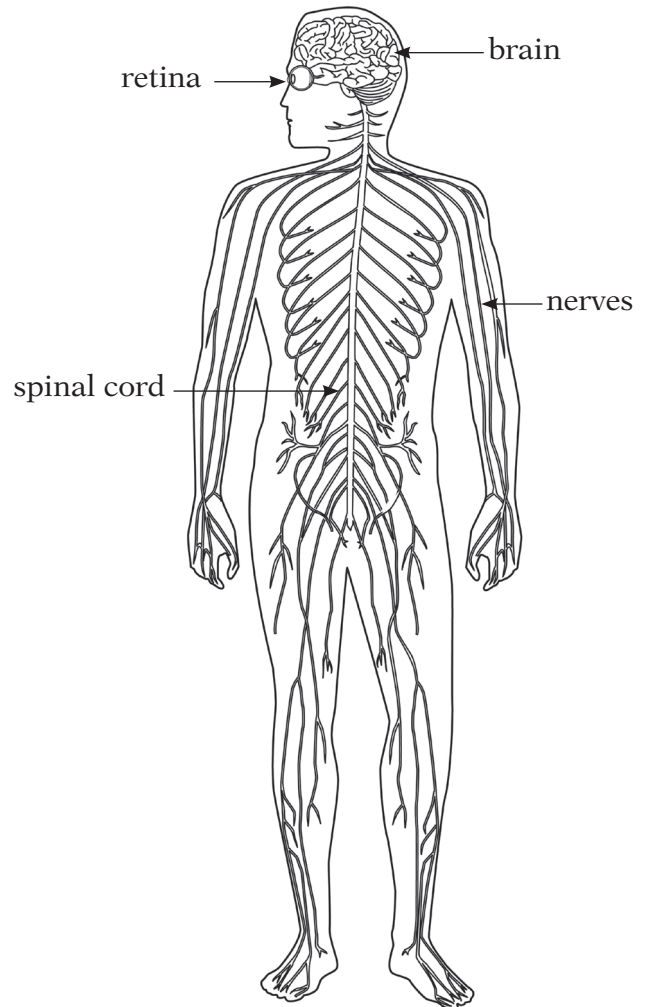
Hint: The labels on the diagram will help you.

In vertebrates (animals with backbones), such as (mahuns) _____, the central nervous system is made up of the (nairb) _____ and the (nalspi) _____ cord. Another important organ is the (nareti) _____ of each eye.

The network of (severn) _____ in the central nervous system is like a system of electrical wiring. The nerves are made up of specialised cells called neurons. Neurons transmit electrical (ssiglan) _____, or messages, along the nerves.

- 2** Using a coloured pen, draw on the diagram the path that an electrical signal would take from the hand to the brain.
- 3** Add the missing vowels (a, e, i, o, u) to complete the following words. Hint: All the words have been used on this page.

- a) nrvs _____
- b) nrns _____
- c) trnsmt _____
- d) lctrcl _____
- e) sgnls _____



The digestive system

BACKGROUND

nutrients (say 'nu-tree-ants') – substances that provide animals and plants with the energy essential for growth and life

The digestive system enables living things to take in and process nutrients to feed themselves and provide energy for their cells.

 **1** Look at the diagram and answer the questions.

Major organs in the human digestive system are:

- a) the ___ o ___ a ___
- b) the ___ a ___ i ___ e ___ i ___ e
- c) the o e ___ o ___ a ___ u ___
- d) the ___ a ___ e i ___ e ___ i ___ e

2 Imagine you are following a tiny piece of food around the human body after it has been eaten. Number these stages of its journey in the correct order.

Step ___: The food is broken down into tiny parts by the acids in the stomach.

Step ___: The food is broken down by saliva in the mouth.

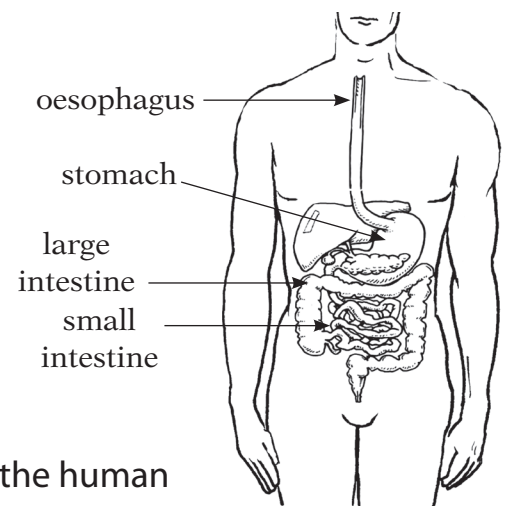
Step ___: The faeces is expelled from the body.

Step ___: The food moves down the oesophagus.

Step ___: The food lands in the stomach.

Step ___: The remaining parts of the food leave the stomach and travel into the small intestine.

Step ___: The food moves into the large intestine, where it is stored as faeces.



Reproduction

All types of living things are able to produce 'offspring' (descendants) that are almost copies of themselves. This process is called **reproduction**. Reproductive systems enable living things to make seeds, lay eggs, have babies or produce offspring in other ways.

-  **1** Read the following passage and then write short answers to the following questions.

In humans, there are two sexes: males and females. Both carry genetic material called DNA in their cells. When one cell from each partner combine, the DNA of the two parents mixes together. The male provides the sperm cell and the female provides the egg cell. The sperm fertilises the egg, entering through the cell membrane and releasing its contents. In this way, the two cells combine. The new cell begins to divide. It grows and develops into a new human.

- a)** What is DNA and where is it found?

- b)** What type of cell does the male parent provide?

- c)** What type of cell does the female parent provide?

- d)** What does the word *fertilises* mean? Use your dictionary if needed.

- 2** Unscramble the words in brackets to complete the passage about DNA. The first letter of each word has been given.

The (ticenge) g_____ material in a cell's (sunclue) n_____ is called (AND) D_____. The DNA works like a set of instructions for building and operating the new human. During (retioncudrop) r_____, the DNA of two (stenpar) p_____ mixes together.

Reproduction CONT'D

The reproductive system

The reproductive system in humans involves sex organs that create sperm and eggs. It also includes physical parts that allow the male and female partners to ensure that the cells meet. These organs include the testes and penis in the male and the ovaries, uterus and vagina in females.

3 Are the following statements true or false?

- | | | |
|---|------|-------|
| a) Reproductive systems enable plants to make seeds. | True | False |
| b) Laying eggs is an important part of reproducing for all animals. | True | False |
| c) In humans, there are two sexes: males and females. | True | False |
| d) Both male and female humans carry DNA in their cells. | True | False |
| e) Female humans provide the sperm cell and males provide the egg. | True | False |
| f) The egg goes through the cell membrane and releases its contents. | True | False |
| g) Cell division causes a new human to grow and develop. | True | False |
| h) Reproductive organs in human males include the testes and penis. | True | False |
| i) Only human females have ovaries, a uterus and a vagina. | True | False |
| j) Animals do not give birth to live babies. | True | False |

4 Select the correct word from the options in brackets. Circle your choices.

Once the sperm cell (fertilises / reproduces) the egg, the new cell begins to (divorce / divide). It forms a ball of cells that burrows into the lining of the (male's / female's) uterus, which is located in the mother's lower body. The cells (multiplex / multiply) thousands and thousands of times. After about 40 weeks, the human baby is fully formed and ready to be (born / brawn).

Check your understanding

Circle the correct answers to these revision questions.

- 1** What helped scientists discover cells, bacteria and viruses?
 - a)** bad spirits
 - b)** diseases
 - c)** the microscope
- 2** What in a cell acts like a little bag full of holes, letting some things through and keeping other things out?
 - a)** the cell membrane
 - b)** the tissues
 - c)** blood
- 3** What are groups of organs called?
 - a)** tissues
 - b)** cells
 - c)** body systems
- 4** The circulatory system pushes around what?
 - a)** the blood
 - b)** the breath
 - c)** skin
- 5** When humans breathe in and out, they
 - a)** absorb carbon dioxide and excrete oxygen.
 - b)** absorb oxygen and excrete carbon dioxide.
 - c)** absorb equal amounts of carbon dioxide and oxygen.
- 6** Where do electrical signals travel?
 - a)** along nerves, to and from the brain
 - b)** along cells, to and from the spinal cord
 - c)** along nutrients, to and from the nerves
- 7** What is the word for when living things make more of themselves, or multiply?
 - a)** combination
 - b)** digestion
 - c)** reproduction
- 8** What does the male partner provide in the process of fertilisation?
 - a)** the egg
 - b)** the sperm
 - c)** the ovaries

What is biological classification?

Scientists use the word **organism** to describe anything that is alive. In science, we can divide types of organisms into different groups or categories. Each group contains organisms that share certain **features** or **characteristics**. Each group can be broken into smaller groups and then into even smaller groups.

There are millions of types of living things. Scientists use groups to classify living things so that they can be specific and exact in the way they study and talk about things.

-  **1** Complete the following passage by writing in the missing letters. Hint: It is the same five letters that are missing throughout.

Biological _____ ification is the process of _____ ifying all living things according to their shared features or characteristics. Scientists _____ ify living things so that they can talk specifically and exactly about them.

- 2** An organism is anything that is alive. List ten examples of organisms.

- 3** Think of two types of plants that you know about and write descriptions of each. Make sure the two plants are very different from each other. Add a sketch for each one.

Name of plant 1: _____

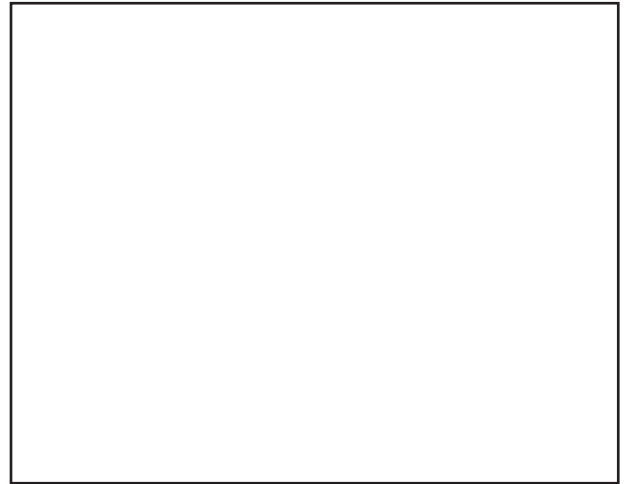
Description:



What is biological classification? CONT'D

Name of plant 2: _____

Description:



4 a) In what ways are the plants in Question 3 similar (alike)?

b) In what ways are they different (unlike)?

5 Think of three different animals. In what ways are they similar to one another?

The hierarchy of biological classification

The biological classification system has **seven ranks** or levels. The groups in the higher levels are larger and broader. They group living things that have just some things in common. The groups in the lower levels are smaller and more specific. They group living things that have many more things in common. The organisation of these groups is called the **hierarchy of classification**. Scientists sometimes disagree with the way others group organisms. For example, some scientists use only five kingdoms.

Level	
Kingdom	There are six kingdoms in the classification system: animal, plant, protista (single-celled organisms including amoebae), fungi, bacteria, archaeobacteria (microorganisms also called extremophiles).
Phylum	There are more than 30 phyla in the Animal Kingdom and 10 in the Plant Kingdom. For example, Phylum Chordata includes humans, birds, fish and all other animals with a backbone.
Class	Each phylum is divided into classes. For example, Phylum Chordata includes the classes of amphibians, birds, mammals, reptiles and fish.
Order	An order is a subcategory of a class.
Family	Families are smaller groups within orders.
Genus	Species that share certain characteristics may be closely related, so they are grouped together in a genus.
Species	A species is a group of living things that can breed to produce fertile offspring.

 **1** Remember the seven levels of biological classification by saying 'King Phillip, Come Over For Good Soup!' Write the seven levels in order from highest to lowest.

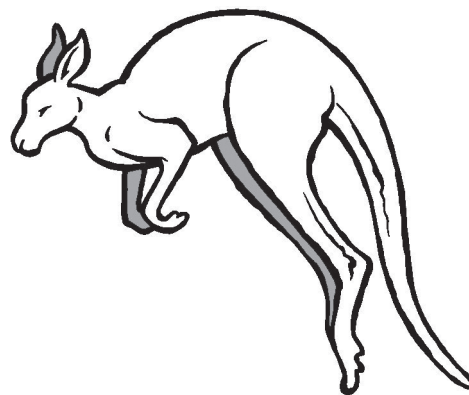
- | | |
|---------|---------|
| 1 _____ | 5 _____ |
| 2 _____ | 6 _____ |
| 3 _____ | 7 _____ |
| 4 _____ | |

The hierarchy of biological classification CONT'D

- Write your own saying to help you remember the seven levels of biological classification. Hint: The first letter of the words in your saying should be, in order, K, P, C, O, F, G and S.

Every living thing can be classified into one group or category in each of the seven ranks. For example, a red kangaroo belongs to the following groups:

- Kingdom: Animal
- Phylum: Chordata
- Class: Mammal
- Order: Diprotodontia
- Family: Macropodidae
- Genus: *Macropus*
- Species: *Macropus rufus*



- Use the information above to fill in the gaps and complete this passage. The first letter of each missing word has been given.

A red kangaroo is grouped in the Animal k_____ – along with birds, fish, humans, monkeys and many other animals – because it has all the characteristics of an a_____ (for example, it is a living organism that feeds on plants or other animals, it can move about and it has a nervous system).

It is also grouped in the Mammal c_____ – along with humans, monkeys and many other mammals – because it has all the characteristics of a m_____ (for example, it is warm-blooded, has fur and feeds its young with milk).

At its lowest or most specific level, the red kangaroo is classified as the s_____ *Macropus rufus*.

Genus and species

BACKGROUND

classification (say 'cla-sif-i-ca-shen') – putting living things into categories so we can be specific and exact in the way we talk about things

We classify living things in groups because they are alike (similar) or unlike (different). Similarities may include the way something moves, what it looks like or how it reproduces. However, animals and plants with the same characteristics or features are only grouped together as a species if they are related genetically. For example, two butterflies may look very much alike, but they may not be closely related. Members of the same species can breed together to produce fertile offspring; if they can't, they are not considered part of the same species.

Writing scientific names

When we write a living thing's scientific name, we include the genus and the species. The genus is the first part of the name and begins with a capital letter and appears in italics. The species comes second and is written in lower case letters and italics.

 **1** Rewrite the following species names correctly.

- a) panthera Leo _____
- b) *equus zebra* _____
- c) Balaenoptera Musculu _____
- d) Homo *sapiens* _____
- e) Macropus *Rufus* _____
- f) camelus bactrianus _____

2 The common name of a living thing is the name that people normally use when talking about it. Can you match these common names with the species names from question 1? Share your answers and try to match them all.

- a) mountain zebra _____
- b) bactrian camel _____
- c) lion _____
- d) blue whale _____
- e) red kangaroo _____
- f) human _____

Unicellular organisms

Some living things are tiny and made up of only one cell. A living thing that is made of only one cell is classified as a unicellular organism. One type of unicellular organism is an **amoeba**. **Bacteria** are also made of only a single cell.

Unicellular organisms are very tiny and most cannot be seen without a microscope. We call them **microorganisms**.

 **1** Fill in the blanks with the correct word from the word bank.

	Word bank
Unicellular organisms are found all over Earth – from the top of mountains to the ocean floor to inside human _____. Some require _____, some do not. Some get food from their _____ and some produce their own food.	oxygen bodies environment
Microorganisms multiply quickly. They usually live in environments that are _____, moist, dark and have plant or _____ material in them. Rubbish bins, compost heaps, _____, clothing and even bedding contain microorganisms.	ponds animal warm
Bacteria and _____ are usually microorganisms. Some types of algae (rootless water plants) and fungi (moulds and yeasts) are _____ too.	amoebae microorganisms
Microorganisms can be _____ or removed with chemicals and are unable to survive in sterile, _____ environments.	clean killed

2 What do you think the prefix 'uni-' means?

Unicellular organisms CONT'D

3 What do you think the prefix 'micro-' means?

4 Are the following words plural or singular?

a) bacteria _____

b) amoeba _____

c) bacterium _____

d) amoebae _____

5 Circle the correctly spelled word in each set of words.

- | | | |
|-------------------|--------------|----------------|
| a) animal | aminal | animel |
| b) ameba | amoba | amoeba |
| c) bateria | bacteria | bacterea |
| d) microorganisms | micorganisms | microorganisms |
| e) microscope | micrascope | microsope |

6 Which of the following environments might contain a lot of live microorganisms?
Circle two choices in each set.

- | | | |
|------------------------------|-------------------|---------------------------|
| a) compost bin | garbage bin | new stainless steel bin |
| b) backyard pond | natural lake | chlorinated swimming pool |
| c) a sterile surgeon's glove | a gardening glove | old woollen gloves |

7 Are the following statements true or false?

- | | | |
|--|------|-------|
| a) Unicellular organisms are found all over the world. | True | False |
| b) All unicellular organisms need oxygen to live. | True | False |
| c) Unicellular organisms are large and easy to see. | True | False |

Multicellular organisms

A multicellular organism is a living thing that is made up of many cells. This category of things has many different organisms in it. Everything from a tiny jellyfish to a human is a multicellular organism – but jellyfish and humans aren't much alike in other ways. Scientists usually break up multicellular organisms into smaller categories using the system of biological classification.

 **1** What does the prefix 'multi-' mean?

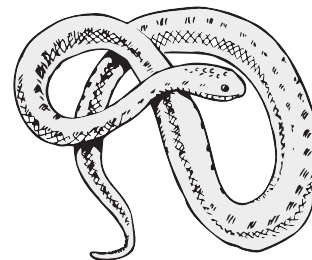
2 Which of these types of organisms are multicellular? Circle them.

- animals • plants • amoebae • bacteria

3 All the organisms listed below are multicellular organisms. Think about what some of these different things have in common. Group them into two or three groups. Write the name of each group, describe the shared features of the group and list each of the organisms.

Hint: There is no right or wrong answer, just different ways of grouping things.

- sea star • earthworm • crab • banksia
- duck • mushroom • snake • sunflower



Group name			
Shared features			
Organisms			

Multicellular organisms CONT'D

- 4 Imagine a tropical rainforest. Look at this list of some multicellular organisms that could be found there. Use the table to organise your list of organisms into the areas of the rainforest where they can be found.

vines	butterflies	turtles	dragonflies
fungi	owls	worms	lichen
short bushes	ferns	bats	grasshoppers
mice	bacteria	alligators	eels
frogs	monkeys	orchids	possums
water snakes	wild turkeys	parrots	moss

On the ground (dark and wet)	Near the ground (shady and green)	High up in the trees (sunny, dry and warm)	In the water and marshes (shallow and cool)

- 5 Compare your answers to Question 4 with a classmate. Which organisms did you categorise differently? Discuss why.
- 6 Give another way that you could group the organisms in Question 4.

Other ways of grouping organisms

There are all sorts of features that can help us organise groups of living things. Organisms can be grouped by biological classification, but they can also be grouped by where they live (**habitat**), what they eat or how they behave. For example, the blue-billed duck (*Oxyura australis*) is grouped with other birds in the Class Aves (Birds), but it could also be grouped with Australian wetland creatures.

Many different types of organisms can live together in the same habitat. Some organisms that share the same living area depend on each other for their survival. For example, organisms that live on coral reefs are **interdependent**, which means that they all need each other to survive. Some organisms are food for others, while some provide shelter or protection to other species.

-  **1** Read the passage and underline all the plants and animals mentioned.

A coral reef is home to many different types of life. Reefs contain lots of different species of fish. These fish eat the seaweed and grasses that grow on the coral. It is important that there is a rich supply of food for all the reef's creatures. Sharks and other predators eat large fish, which in turn eat smaller fish. All sorts of plants and animals live in tropical reef habitats. They include eels, crabs, prawns, sea stars, seahorses, plankton, seaweed, anemones, clownfish, angelfish, coral polyps, sea turtles, jellyfish, dolphins, dugongs and whales.

- 2** How many of these living things can be classified in the Animal Kingdom? List them here. Hint: List them only once each, even if they are mentioned more than once in the passage.

Other ways of grouping organisms CONT'D

3 Which three things can be classified in the Plant Kingdom?

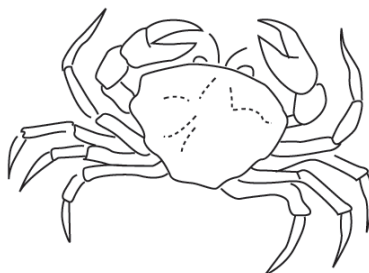
4 List any that you are unsure about.

5 Circle the odd one out in each group of living creatures.

- | | | |
|---------------|-----------|------------|
| a) fish | plankton | shark |
| b) seaweed | anemone | dolphin |
| c) turtle | crab | jellyfish |
| d) sea star | angelfish | large fish |
| e) clown fish | seahorse | horse |

6 For each of your answers in Question 5, list the difference or differences that make that organism the odd one out.

- a) _____
- b) _____
- c) _____
- d) _____
- e) _____



Check your understanding

Circle the correct answers to these revision questions.

- 1** The classification system helps us
 - a)** see the cells of tiny creatures.
 - b)** be very exact in the way we describe living things.
 - c)** describe things to animals.
- 2** The saying 'King Phillip, Come Over For Good Soup' represents these categories:
 - a)** Kingdom, Phillip, Country, Order, Family, Generation, Species.
 - b)** Kingdom, Phylum, Class, Order, Family, Genus, Species.
 - c)** Kingdom, Phylum, Class, Order, Family, Genius, Species.
- 3** One type of living thing that has only one cell is called
 - a)** an amoeba.
 - b)** a microorganism.
 - c)** an animal.
- 4** Humans and jellyfish are
 - a)** very alike.
 - b)** both unicellular organisms.
 - c)** both multicellular organisms.
- 5** One of the six kingdoms of classification is
 - a)** jellyfish.
 - b)** fungi.
 - c)** rocks.
- 6** A coral reef, which is home to many living things, contains
 - a)** both plants and marine animals.
 - b)** just plants.
 - c)** just marine animals.
- 7** We can group living things in different ways. Some ways include
 - a)** by their movements and their habitats.
 - b)** by the colour of their fur.
 - c)** by the number of them living in the wild.
- 8** Phylum Chordata includes these classes:
 - a)** amphibians, animals, reptiles and turtles.
 - b)** amphibians, birds, mammals, reptiles and fish.
 - c)** amoeba, bacteria, mammals, reptiles and fish.

How do plants get food?

BACKGROUND

nutrients (say 'nu-tree-ants') – substances that provide animals and plants with the energy essential for growth and life

Just like animals, plants need food to produce energy, so they can grow and survive. Plants have two main types of food: organic and inorganic. Organic means coming from living things, and inorganic means coming from things that are not living.

Inorganic foods are the minerals that plants get from the earth. They are nutrients that are found in the soil that the plants grow in. Minerals in the soil are dissolved in water and sucked up (absorbed) by the plant's roots. In areas where the soil is poor quality – either with too few nutrients or too many of one kind of nutrient – humans can add fertilisers to give plants more inorganic nutrients.

Green plants can also make their own **organic food**, using sunlight and two important inorganic nutrients: carbon dioxide and water. The name for the organic food that plants make out of inorganic nutrients is glucose. The process of making glucose is called photosynthesis.

 **1** Why might plants need fertiliser? Complete the sentence.

Plants sometimes need fertiliser because _____
_____.

2 Fill in the gaps using words from the word bank.

a) Plants take in _____ from the soil
when they _____ water from the soil.
This process is called absorption.

b) Plants produce _____ from
_____, sunlight and water.
This process is called photosynthesis.

Word bank

absorb
carbon dioxide
glucose
minerals

How do plants get food? CONT'D

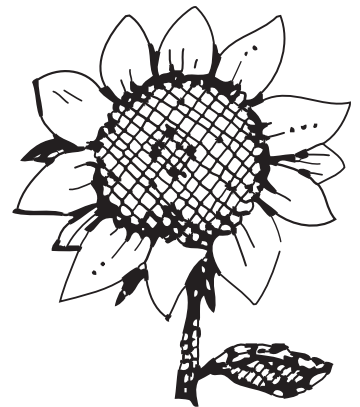
3 Mark each of the following things as organic or inorganic.

- | | | |
|--------------------------|----------------------------------|------------------------------------|
| a) plants | ☐ organic | ☐ inorganic |
| b) carbon dioxide | ☐ organic | ☐ inorganic |
| c) glucose | ☐ organic | ☐ inorganic |
| d) soil | ☐ organic | ☐ inorganic |
| e) water | ☐ organic | ☐ inorganic |
| f) sunlight | ☐ organic | ☐ inorganic |
| g) oxygen | ☐ organic | ☐ inorganic |

4 a) What are the two inorganic nutrients that green plants need to make the organic food called glucose?

b) Green plants turn these ingredients into what?

5 In your own words, write two reasons why plants need food.



Inside plant cells

BACKGROUND

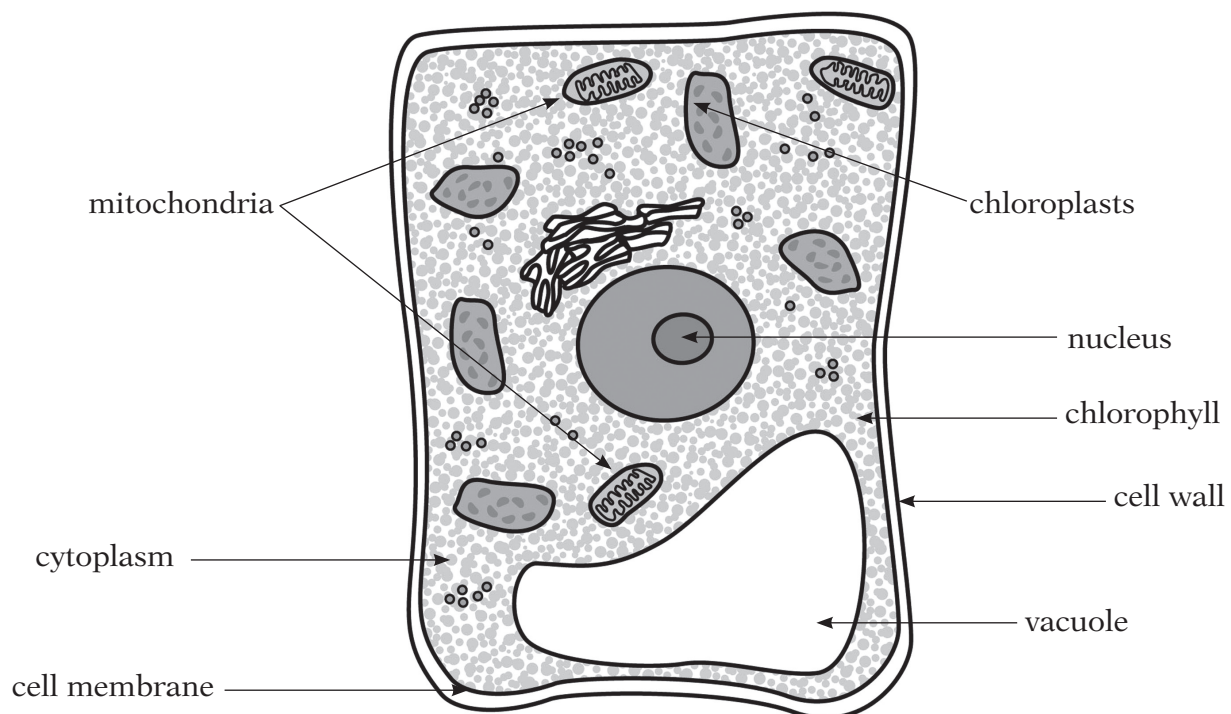
organelles (say 'or-gan-elles') – tiny living machines that are found inside cells

- 1 Complete the following passages by filling in the missing vowels (a, e, i, o, u).

Hint: The labels on the diagram will help you.

Plant cells have a c__ll m__mbr__n__ that is soft. This m__mbr__n__ is surrounded by a protective layer called the c__ll w__ll. The c__ll w__ll is made of a substance called cellulose. It is what gives plants their rigid form and shape.

The parts of a plant cell are floating in a jelly-like substance called cyt__pl__sm. A plant cell has a n__cl__ __s, which controls the cell. Also inside the cell are organelles called chl__r__pl__sts. These contain chl__r__phyll, a green substance that plants use to make their food. There are m__t__ch__ndr__ __ too, which are organelles that help provide the cell with energy. There are also areas called v__c__ __les, which store water inside the cell.



- 2 In your own words, write descriptions of each of these parts of plant cells. Use complete sentences.

a) cell membrane

Inside plant cells CONT'D

b) cell wall

c) chloroplast

d) chlorophyll

e) cytoplasm

f) mitochondria

g) nucleus

h) vacuole

Do plants breathe?

Plants take in air, like humans do, although we don't call it 'breathing'. Most plants have leaves, which are made up of layers of cells. In the leaves are tiny holes. These holes can open and close to let gases in and out. Plants do this during two processes: **photosynthesis** and **respiration**. Plants respire all the time, but they only photosynthesise in sunlight.

Word bank

dioxon carbide
muhans
oselugc
gyoxen

-  **1** Unscramble the words in the word bank and then write them in the answer spaces to complete this passage about photosynthesis.

Plants take in _____ gas and release oxygen gas. Carbon dioxide is the gas that _____ and animals breathe out. Plants use this gas, along with water and sunlight, to produce the food _____. A side product of this process is _____ – the most important gas in the air for humans. This process is called photosynthesis. The word equation for this is:

carbon dioxide + water (+ light energy) = glucose + oxygen

- 2** Fill in the vowels (a, e, i, o, u) to complete this passage about respiration.

Like animal cells, plant cells need __xyg__n to survive. The plant cells need __xyg__n to help them release energy from gl__c__s__. The plant takes in __xyg__n and releases c__rb__n d__ __x__d__ as a waste product. This process is called respiration. The word equation for this is:

glucose + oxygen = carbon dioxide + water (+ energy)



Photosynthesis

Photosynthesis is the name we give to a special process that green plants use to make food using sunlight, or **light energy**. They use light energy to change **carbon dioxide** and **water** into a special sugar called **glucose**. Plants get their energy from glucose, and this energy allows them to grow and carry out all sorts of functions.

Green plant cells contain a green chemical called **chlorophyll**, which traps light energy from the sun. During photosynthesis, water molecules are split apart using this light energy. The hydrogen parts of each water molecule join with carbon dioxide to make glucose, the special sugar that plants have as their food. The leftover oxygen part of each water molecule is released by the plant through its leaves.

The word photosynthesis is made up of two parts – *photo* means ‘light’ and *synthesis* means ‘to make something’.

 **1** Green plants use sunlight to turn what two things into food? The first letter has been given for each.

w _____

c _____

2 Green plants produce what two things as a result of photosynthesis? The first letter has been given for each.

g _____

o _____

3 Which of the things in Question 2 is used by the plant as food?

4 Which of the things in Question 2 is released as a waste product?

5 Write a word equation for what happens in photosynthesis. Use your answers from Questions 1 and 2, plus the ingredient sunlight. Use ‘+’ and ‘=’ symbols.

Photosynthesis CONT'D

Chlorophyll and sunlight

Sunlight is a form of energy. The sunlight we see is actually made up of different wavelengths of light, which create three different colours – green, red and blue.

The chlorophyll in the cells in a plant's leaves absorb the energy from blue light and red light. They reflect green light back without absorbing it. This is why plants look green to us. They are reflecting light from the green range of the spectrum back at us.

6 Write the colours of light into the correct spaces in the table.

red light blue light green light

Colour of light	Usefulness to plants
	reflected back without being absorbed
	absorbed by the plant's leaves
	absorbed by the plant's leaves

7 Are the following statements true or false?

- | | | |
|---|------|-------|
| a) During photosynthesis, water molecules are split apart. | True | False |
| b) Plants use light energy from the sun to make carbon dioxide. | True | False |
| c) The oxygen part of water is released as a gas. | True | False |
| d) The hydrogen parts of water join with carbon dioxide to make glucose. | True | False |
| e) Glucose is a mineral from the soil that plants use as their food. | True | False |
| f) Glucose is a special sugar made during photosynthesis. | True | False |

Plant structures

Most green plants have some common structures. These structures play different parts in the process of making food (photosynthesis) and distributing this food around the plant.

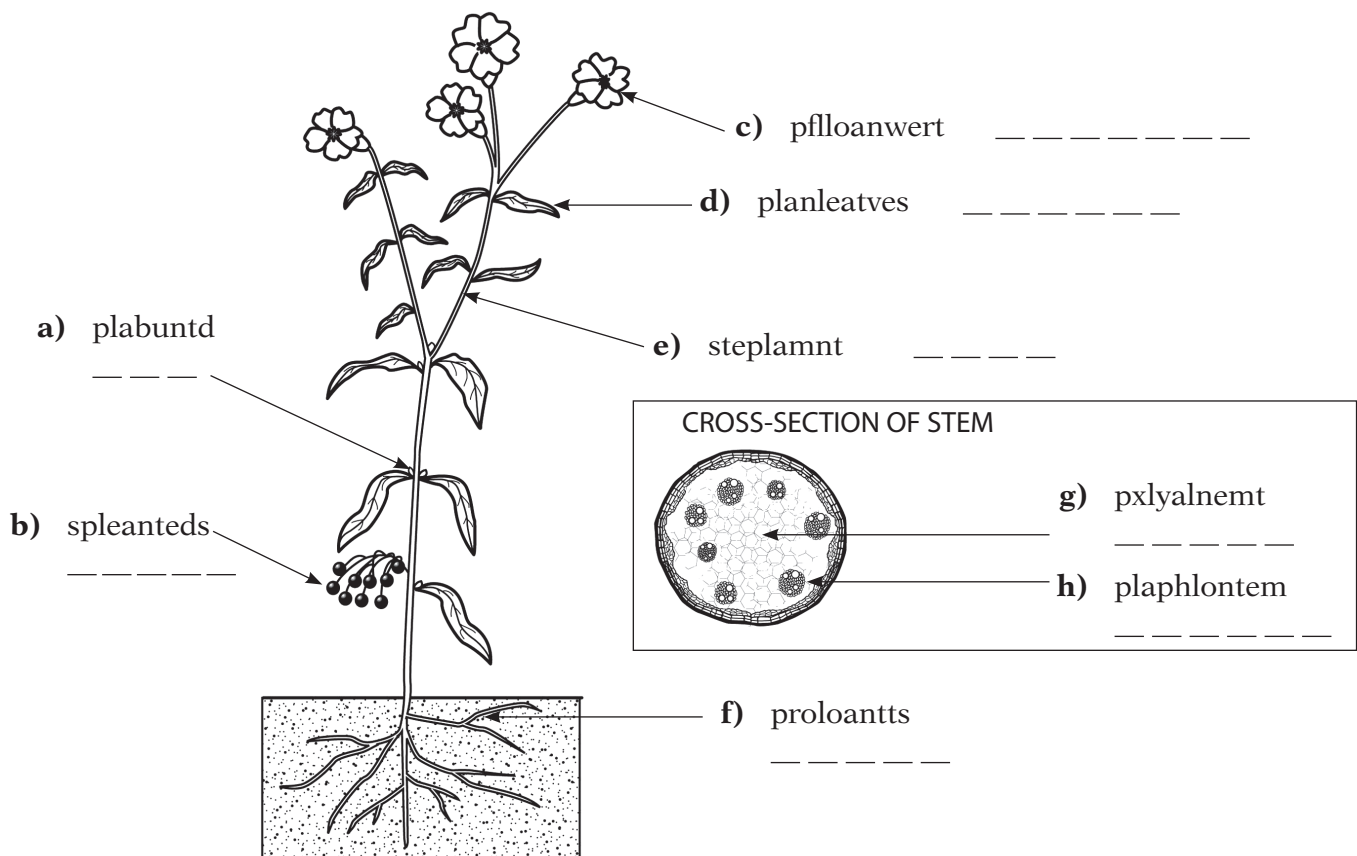
In most flowering plants, there are roots below the ground. They keep the plant in place by holding on to the soil. Roots absorb water for the plant to use in photosynthesis.

Cells in the plant's leaves carry out the work of photosynthesis, creating food (glucose) by absorbing sunlight and using water and carbon dioxide.

Many flowering plants have a system of tubes inside their stems. Two of these tubes are called the xylem and the phloem.

- The **xylem** carries water from the plant's roots around to all the plant's cells.
- The **phloem** carries food (glucose) from the leaves to the rest of the plant. This gives each cell the energy it needs to survive and grow.

 **1** Extra letters have been added to these eight words, which label the parts of the plant. In each case, the extra letters spell *p l a n t*. Cross out the letters p, l, a, n and t from each word puzzle in order.



Check your understanding

Circle the correct answers to these revision questions.

- 1** Plants need food to survive. What is the sugary substance they make?
 - a)** honey
 - b)** glucose
 - c)** phloem
- 2** Carbon dioxide is
 - a)** a gas.
 - b)** an inorganic nutrient.
 - c)** both a gas and an inorganic nutrient.
- 3** Plant cells contain
 - a)** mitochondria and red blood cells.
 - b)** a nucleus and cytoplasm.
 - c)** a blue substance called chloroform.
- 4** Plants can take in carbon dioxide and give out oxygen. This helps humans because
 - a)** we breathe in carbon dioxide.
 - b)** we need more carbon dioxide.
 - c)** we breathe in oxygen.
- 5** In photosynthesis, water molecules are broken apart. Water molecules contain
 - a)** hydrogen and oxygen.
 - b)** just hydrogen.
 - c)** carbon dioxide.
- 6** Plants reflect
 - a)** blue light.
 - b)** green light.
 - c)** red light.
- 7** Chlorophyll is stored in
 - a)** chlorofillers.
 - b)** chloroplasts.
 - c)** vacuoles.
- 8** The xylem and the phloem carry
 - a)** water and glucose through the stem.
 - b)** water and oxygen through the roots.
 - c)** glucose and seeds to the leaves.

The importance of blood

 **1** Read the following information about blood and then answer the questions.

All animals, including humans, need certain things if they are to stay healthy and survive. Some things that are vital to life are water, oxygen, nutrients from food and a way of getting rid of waste products. All animals must have a way of transporting water, oxygen, nutrients and waste products around the body, to and from every single cell. In vertebrates (animals with a backbone), these substances are carried in the **bloodstream**.

Blood is made up of red blood cells, white blood cells and platelets travelling around the body in a fluid called plasma.

- **Red blood cells** carry oxygen to all the cells in the body.
- **White blood cells** help protect the body from infection.
- **Platelets** are involved in blood-clotting.
- **Plasma** carries the blood cells, as well as nutrients and waste products. It is mostly made up of water.

The network of vessels that carry blood is called the **circulatory system**. When blood travels around the body, we call it circulation. The heart is the organ that pumps the blood around the circulatory system.

a) What is the network of vessels that carry an animal's blood called?

b) What four vital things does an animal's body need to stay healthy?

c) Which part of blood carries oxygen to the cells? _____

d) Which part of blood carries nutrients to the cells? _____

e) Which part of blood carries away waste products from the cells?

The importance of blood CONT'D

2 Correct the spelling of these phrases by rewriting them in the spaces provided.

- a) the sells of anamals _____
- b) stay helthy _____
- c) have enuf enegy _____
- d) oxygin is in the blud _____
- e) the sircultary sistem _____

3 Use the words in the word bank to complete these facts.

- a) About 60 per cent of the adult human body is made up of _____.
- b) Fish are vertebrates that take in _____ through their gills.
- c) Carbon dioxide and lactic acid are two types of _____ transported by the circulatory system.
- d) _____ such as glucose are carried by plasma.

Word bank

oxygen
waste products
water
nutrients

4 Complete the table to show that you understand the word forms.

- *circulation* (a noun or thing)
- *circulate* (a verb or action)
- *circulatory* (an adjective or description)

Word form	Meaning
a)	to go around
b)	the process of blood circulating
c)	the name that describes a vertebrate's blood system

Turning food into energy

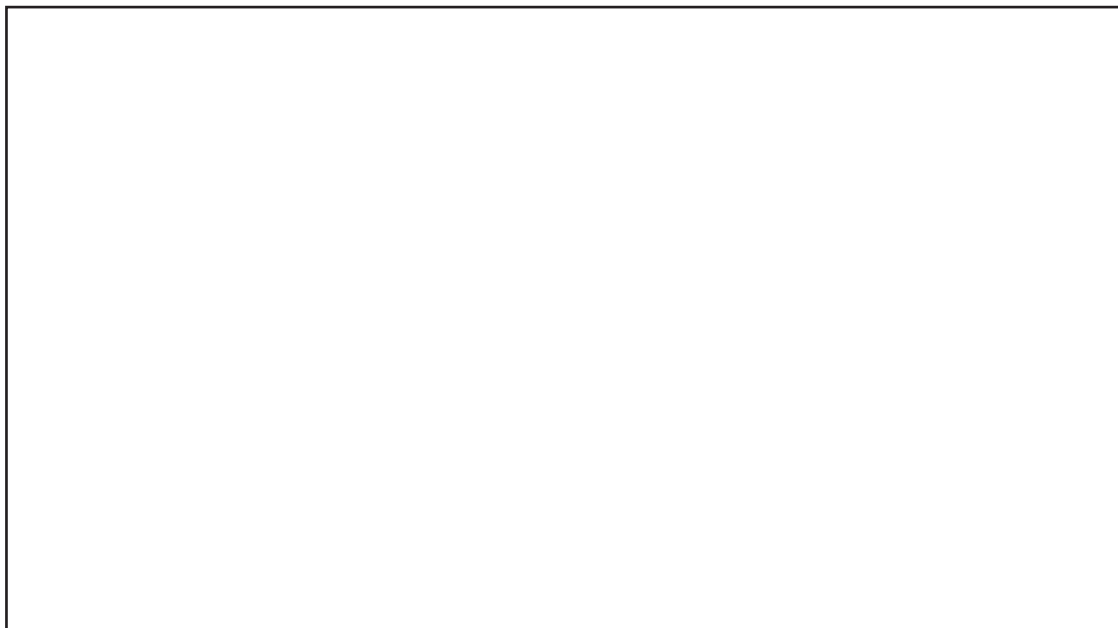
-  **1** Cross out the incorrect word from each pair in the brackets to complete the passage.

Living things need (sugar / energy) to stay alive. When an animal eats food, its (digestive / circulatory) system breaks the food down into smaller bits called nutrients. These nutrients are picked up by the (atmosphere / bloodstream) and transported around the animal's body to all the different cells. When a substance goes into the bloodstream we call this (absorption / reproduction).

The digestive system breaks down parts of foods called carbohydrates into a type of sugar called (oxygen / glucose). This is a sugar that gives (humans / minerals) and animals their energy needs.

- 2** Draw a flow diagram that shows what happens to food when it is eaten by an animal. Label the diagram with the letters a–f to represent these steps.

- the food is swallowed
- the food is broken into smaller bits
- the nutrients enter the bloodstream
- the nutrients are taken to the cells
- the nutrients are absorbed by the cells
- the cells release waste products



Turning food into energy CONT'D

Giving cells glucose is like filling a car up with fuel. Glucose gets things moving, providing the animal with energy so that its cells can perform their functions to keep themselves alive and healthy.

3 Complete the answers to these step-by-step questions.

- a) What tiny bits are all living things made of? c _ _ _ s
- b) How do living things get energy? By taking in f _ _ _
- c) What system involves taking in food? The d _ _ _ _ _ _ _ _ _ _ system
- d) What sugary substance does the body make food into? g _ _ _ _ _ _ _
- e) How is this substance taken around the body? Through the
c _ _ _ _ _ _ _ _ _ _ y s _ _ _ _ m
- f) Why do animals eat? To provide all the body's c _ _ _ _ s with energy to live and grow

4 Rearrange these fragments to write proper sentences.

- a) to survive Animals need energy and other living things
_____.
- b) The blood in the circulatory system all the body's cells carries glucose to
_____.
- c) part in the Glucose plays an important digestive system
_____.

5 Glucose contains energy. List two things that humans need energy to do.

Waste systems

As well as bringing good things to the cells, the circulatory system also takes away bad things. After cells have used up the energy that they release from glucose, certain waste products are left behind. These can become poisonous if they remain in the cells and build up. So the cells release these waste products.

Blood cells take the waste products away to other organs that are designed to deal with waste safely. The kidneys, liver, lungs and skin are some of the organs involved in waste removal.

 **1** Write these phrases in the correct places in the table.

- get their energy from glucose
- and deliver waste products to the organs to be dealt with safely
- and takes away bad things from the cells

The bloodstream	brings good things to the cells	
Cells		and release waste products from glucose.
Blood cells	take nutrients to the cells that make up the organs	

2 Using your own knowledge, match these waste products with the organs that help remove them from the body.

Waste products	Forms	Organs
salts and water	urine and sweat	a) b)
carbon dioxide and water vapour	as gas that is breathed out	c)

Check your understanding

Circle the correct answers to these revision questions.

- 1** Animals need food (nutrients), water and oxygen to survive. These substances are carried around the body by the
 - a)** skin cells.
 - b)** lungs.
 - c)** blood.
- 2** The bloodstream moves around a vertebrate's body through
 - a)** the arms.
 - b)** a network of vessels.
 - c)** electricity.
- 3** The name we give to a vertebrate's blood system is
 - a)** the digestive system.
 - b)** the circulatory system.
 - c)** the central nervous system.
- 4** Human lungs help remove waste
 - a)** gases.
 - b)** solids.
 - c)** blood.
- 5** Red blood cells carry
 - a)** vitamins.
 - b)** hydrogen.
 - c)** oxygen.
- 6** Each body cell needs to be able to take in
 - a)** nutrients.
 - b)** air.
 - c)** dead blood cells.
- 7** Each body cell needs to get rid of
 - a)** circulation.
 - b)** kidneys.
 - c)** waste products.
- 8** Three organs that are important to waste removal are
 - a)** the skin, the piano and the organ.
 - b)** the kidneys, the liver and the heart.
 - c)** the kidneys, the liver and the lungs.

Types of nutrients

BACKGROUND

nutrients (say 'nu-tree-ants') – substances that provide animals and plants with the energy essential for growth and life

When we eat food, our bodies use the different substances found in the food in different ways. These substances are known as **nutrients**. Human bodies need the right kinds of nutrients in the right amounts to stay healthy. The digestive system processes the food we eat. Digested nutrients are absorbed through the walls of the small intestine and then transported to all the cells through the blood in the circulatory system.

There are seven types of nutrients:

- protein
- vitamins
- minerals
- carbohydrates (which are converted into glucose)
- fibre
- fats
- water.

 **1** Find the words in the introduction that have an opposite meaning to the words listed below.

- a) same _____
- b) unhealthy _____
- c) wrong _____
- d) large _____
- e) unknown _____

2 Write 'circulatory' or 'digestive' to complete these sentences.

- a) The food we eat is processed by the _____ system.
- b) Blood transports nutrients to the cells via the _____ system.
- c) Blood is one part of the _____ system.
- d) The small intestine is one part of the _____ system.

Types of nutrients CONT'D

Vitamins are found in natural foods such as fruit and vegetables. The body needs vitamins to do all sorts of different jobs.

- Vitamin A helps our eyesight.
- Vitamin C keeps our immune system strong.
- Vitamin E helps keep our skin healthy.
- Vitamin D repairs our bones (and the human body can also make it from sunlight).



Humans also need **minerals**, which are found in natural foods and come from the soil and water. Examples of minerals that the body uses are iron, sodium and calcium.

3 Write the letters of the vitamins next to the jobs they do in our bodies.

- a) Vitamin ____ = keeps skin healthy
- b) Vitamin ____ = boosts immune system
- c) Vitamin ____ = helps repair our bones
- d) Vitamin ____ = helps eyesight

Humans need **fibre**, which is the stuff plant cell walls are made of. Humans also need to eat **protein** and certain **fats** in their diet to stay healthy.

4 Cross out the incorrect word from each set of brackets to complete these passages.

The human body needs (artificial / natural) foods such as fruit and vegetables. Vitamins, minerals and fibre come (form / from) these foods. Fruit and vegetables get their minerals from (soil / soiled) and water.

Humans also need (fibre / fibrous), which is the stuff plant cell walls are made of. The hard cell walls of plants are made of (cellulose / cellular), a fibre that we eat when we have (fruity / fruit) and vegetables.

Humans get protein and certain (fast / fats) from meats, nuts, seeds, legumes, fish and eggs.

Types of nutrients CONT'D

5 For each type of nutrient listed, give at least three examples of foods that contain that nutrient.

- a) carbohydrates _____
- b) protein _____
- c) vitamins _____
- d) minerals _____
- e) fibre _____
- f) fats _____
- g) water _____

6 Each row of foods contains one item that doesn't belong. Circle the item and write why it doesn't belong with the others. Hint: Think about the type of nutrients each one contains.

- a) eggs red meat banana chicken fish prawns

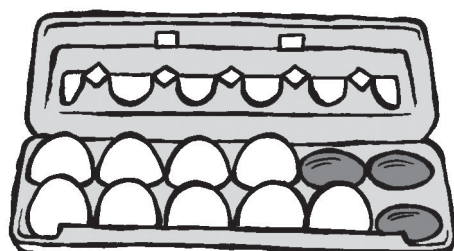
Why it doesn't belong: _____

- b) broccoli lettuce cabbage spinach potatoes parsley cauliflower

Why it doesn't belong: _____

- c) bread butter corn cereal cracker biscuits bran muffin rice pasta

Why it doesn't belong: _____



Body systems working together

The human body contains a number of important organs. Each organ is made up of millions of cells that have very specialised functions. For example, the heart is an organ that pumps blood around the body. Heart cells function quite differently from the cells of other organs, such as lung cells. Lung cells have special parts that help them collect oxygen for red blood cells. Lungs enable humans to breathe air in and provide the oxygen that cells need to survive. In the central nervous system, nerve cells send electric signals to another important organ – the brain. Brain cells are very delicate and control other systems in the body. The spinal cord is part of the system that involves electrical signals being passed to and from the brain.

 **1** Fill in the gaps in the table. The first has been done for you.

Word	Meaning
organ	a structure that has a special job in the body
heart	
	a pair of organs that collect oxygen
nerve cells	
	the gas humans need to keep their cells working

2 Write in the words that correctly complete each sentence.

- In the body, the _____ works by pumping _____ around the body to all the _____.
- In the central nervous system, _____ send electric signals to the brain.
- Our _____ work by collecting oxygen and working with the _____ blood cells to send it to all our cells.

Body systems working together CONT'D

3 Complete the table by writing in the correct letters.

One (singular)	More than one (plural)
lung	_____
n _____ c _____	nerve cells
body	_____

4 a) In the box below, draw the shape of a human body.

b) Mark the positions of the following organs on the body:

- lungs
- heart
- stomach
- brain
- voice box
- skin.

Removing waste products

Our kidneys, liver, lungs and skin all help our bodies to remove waste products that aren't needed. These organs are part of a group of organs called the **excretory system**. To excrete something means to remove it from the bloodstream.

Waste products include carbon dioxide gas and certain types of body waste. Humans excrete liquid waste products through sweat and urine.

 **1** Read these sentences about waste products and then fill in the gaps to rewrite the information in a different order. For example, *When we digest food, waste products are made* can be rearranged and changed to *Waste products are made when we digest food*.

a) Our kidneys make urine out of our waste products and water.

Urine is made from _____ and _____ by our kidneys.

b) Sweat is waste water that comes out of our skin.

Our skin releases _____ that comes out as _____.

c) The excretory system involves different organs that help us remove waste from the body.

The different organs that help us remove waste from the body are part of the _____.

2 Complete the following sentences by filling in the missing vowels (a, e, i, o, u).

a) The waste product c__rb__n d__ __x__d__ leaves the human body every time a person exhales, or br__ __th__s __ __t through their l__ngs.

b) The l__v__r changes toxic ammonia in the blood to the waste product urea. The k__dn__ys filter the urea from the blood and turn it into __r__n__.

c) Waste products are removed through sw__ __t, which comes out tiny openings in the sk__n, called p__r__s.

Removing waste products CONT'D

3 Unscramble these names of waste products:

a) corban idoxide = _____

b) tawer = _____

c) reau = _____

4 Choose the correct words from the word bank to answer these questions.

Word bank

remove

urine

lungs

assist

a) Which word means 'take away'? _____

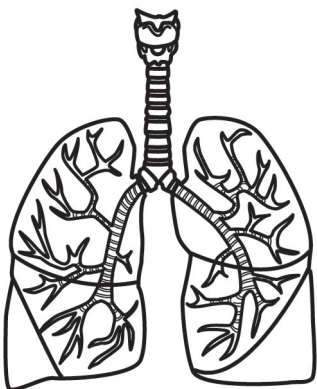
b) Which word means 'help'? _____

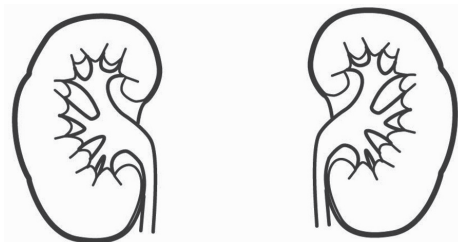
c) Which word means 'breathing organs'? _____

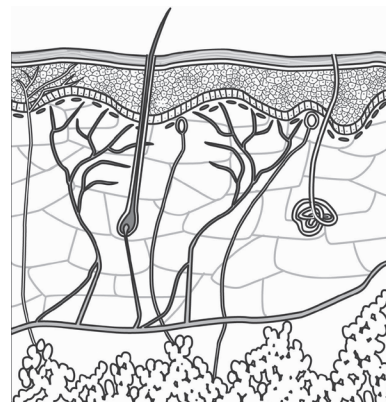
d) Which word means 'a liquid waste product'? _____

e) Write a sentence about the excretory system using all four words from the word bank.

5 Name each organ and write the waste product associated with it.







What else are bodies made of?

Animals and humans are made of cells, which make up tissues, which make up organs, which work together in groups to make up body systems. The bodies of humans and some other animals are built around a structure called a **skeleton**. We call all the bones and joints in the body the **skeletal system**. The bones of our bodies help us stay upright and give us proper shape and strength. Plants do not need skeletons because they have hard cell walls, which give their tissues a solid frame.

Attached to the human skeleton are **muscles**. Muscles allow us to move. Muscles are attached to bones by tissues called **tendons**. Muscles work together to push and pull on our bones, giving us the ability to move. When a muscle pulls, we say it contracts. When the muscle stops pulling, we say it is relaxed. Bones are joined together at places in the body called joints. Knees and elbows are examples of joints. Inside our joints, strings called **ligaments** tie the bones together.

 **1** Read the text below. Circle any words that you think are spelled incorrectly.

We call all the bones and joynts in the body the sceletal system. Mussels are attached to bones by tendins. Muscles work together, often in pears – for example, the biceps and triceps in the arm. When the biceps muscle contracts, the triceps muscle relaxes and the elbow bends. When the triceps contracts, the biceps relaxes and the arm stratens.

2 Rewrite correctly the words that you circled in Question 1.

_____	_____
_____	_____
_____	_____

Check your understanding

Circle the correct answers to these revision questions.

- 1 Humans get substances called vitamins from
 - a) fruit and vegetables.
 - b) rocks.
 - c) cells.
- 2 The vitamin that our skin produces from sunlight is
 - a) Vitamin A.
 - b) Vitamin D.
 - c) Vitamin E.
- 3 Which structures carry oxygen in our bloodstream?
 - a) lungs
 - b) nerve cells
 - c) red blood cells
- 4 The type of energy that helps our nerve cells send signals to the brain is
 - a) electrical energy.
 - b) magnetism.
 - c) gas.
- 5 Which gas is a human waste product?
 - a) oxygen
 - b) helium
 - c) carbon dioxide
- 6 The removal of waste products from the body is called
 - a) execution.
 - b) excretion.
 - c) nervous system.
- 7 The human body has a structure of bones and joints called a
 - a) cell.
 - b) skeleton.
 - c) tissue.
- 8 Bones and muscles are attached by tissues called
 - a) joints.
 - b) tendons.
 - c) ligaments.

1 Cells

What are cells? 1 a, b, e. 2 a, b, e. 3 a tiny parcels of matter; building blocks b different functions or jobs. 4 blood cells, nerve cells, skin cells; extra answers could include stem cells, neurons, heart-muscle cells. 5 a liv/ing b or/gan/is/m c cell/u/lar d u/nits. 6 a living, many, cells b one, cell, organism. **The structure of cells** 1 nucleus, cell membrane, cytoplasm. 2 cells, cell, cells, cell wall, cellulose, cell walls, tall, cells, cell walls. 3 a cells b bodies c soft. 4 cellulose. **Animal cells** 1 a cell membrane, ribosomes, mitochondria, vacuole, nucleus, Golgi body. 2 blood cells, nerve cells, skin cells. **Plant cells** 1 a cell membrane, cytoplasm, nucleus, vacuole, chloroplast, ribosomes, Golgi body. 2 chlorophyll, chloroplast. **What do cells do?** 1 jobs, healthy, grow, nucleus, nutrients, energy, plant, water, animals, food, body, waste. 2 a healthy, growing, living b sunlight, water, nutrients. **Cell division** 1 a divide b divides c dividing d divided e division. 2 themselves, replace, create, cell, two, identical. 3 Other names for mitosis are cell division and cell reproduction. **Check your understanding** 1 b. 2 a. 3 a. 4 b. 5 b. 6 c. 7 c. 8 a.

2 Body systems and cells

How do we know about cells? 1 a 1663 b 1595 c 1838–39. 2 a microscope b Sample answers: microorganism, microchip, microclimate, microwave, microphone, microsurgery. 3 Sample answers: a small b sicknesses c made-up d strong e function f well. 4 a parcel b bacteria c disease d microscope. 5 Answers will vary. **What jobs do cells do in humans?** 1 a oxygen b jobs c membrane d cells e microscope. 2 a. 3 b. 4 Red blood cells carry oxygen to the rest of the body's cells. The cell membrane lets in oxygen, and a substance called haemoglobin holds on to it tightly. When the red blood cell reaches other cells, it delivers the oxygen through the cell membrane. 5 oxygen. 6 cellular, bodily, gaseous, microscopic. **Tissues, organs and body systems** 1 Sample answers: heart – pumps blood around the body; lungs – collect oxygen for the bloodstream; skin – protects the body and excretes waste water. 2 Systems are made up of organs. 3 Sample answers: brain – nervous system, stomach – digestive system, trachea – respiratory system. 4 brain, skin, lungs, trachea, heart, stomach. **Body systems** 1 a the system that passes electrical signals to and from the brain b the system that transports blood around the body c the system that deals with the taking in and processing of food d the system that includes the sex organs e the system that includes the organs used for

breathing, or respiration f the system that removes waste products from the body g the system of bones, cartilage, ligaments and tendons. 2 Answers will vary. **The circulatory system** 1 different, same, humans, oxygen, body, simple, complex, larger, organ, blood, cells. 2 Sample answer: Body systems are interdependent. This means that each of the body systems relies on the other body systems to function properly. **The central nervous system** 1 humans, brain, spinal, retina, nerves, signals. 3 a nerves or nervous b neurons c transmit d electrical e signals. **The digestive system** 1 a the stomach b the small intestine c the oesophagus d the large intestine. 2 4, 1, 7, 2, 3, 5, 6. **Reproduction** 1 a DNA is genetic material found in the nucleus of cells. b sperm cell c egg cell d cause to develop a new individual. 2 genetic, nucleus, DNA, reproduction, parents. 3 a true b false c true d true e false f false g true h true i false j false. 4 fertilises, divide, female's, multiply, born. **Check your understanding** 1 c. 2 a. 3 c. 4 a. 5 b. 6 a. 7 c. 8 b.

3 Classifying living things

What is biological classification? 1 classification, classifying, classify. 2 Sample answers: humans, dogs, monkeys, fish, plants, bacteria, fungi, bugs, turtles, algae. 3, 4, 5 Answers will vary. **The hierarchy of biological classification** 1 kingdom, phylum, class, order, family, genus, species. 2 Answers will vary. 3 kingdom, animal, class, mammal, species. **Genus and species** 1 a *Panthera leo* b *Equus zebra* c *Balaenoptera musculus* d *Homo sapiens* e *Macropus rufus* f *Camelus bactrianus*. 2 a *Equus zebra* b *Camelus bactrianus* c *Panthera leo* d *Balaenoptera musculus* e *Macropus rufus* f *Homo sapiens*. **Unicellular organisms** 1 bodies, oxygen, environment, warm, animal, ponds, amoebae, microorganisms, killed, clean. 2 one. 3 small. 4 a plural b singular c singular d plural. 5 a animal b amoeba c bacteria d microorganisms e microscope. 6 a compost bin, garbage bin b backyard pond, natural lake c a gardening glove, old woollen gloves. 7 a true b false c false. **Multicellular organisms** 1 many. 2 animals, plants. 3 Answers will vary. 4 Sample answer: On the ground: fungi, mice, bacteria, worms, moss, lichen. Near the ground: vines, short bushes, ferns, wild turkeys, orchids, grasshoppers. High up in the trees: butterflies, owls, monkeys, bats, parrots, possums. In the water and marshes: frogs, water snakes, turtles, alligators, dragonflies, eels. 6 Sample answer: biological classification (e.g. animals, plants, bacteria, fungi). **Other ways of grouping organisms** 1 coral, fish, seaweed, grasses, sharks, large fish,

smaller fish, eels, crabs, prawns, sea stars, seahorses, plankton, seaweed, anemones, clownfish, angelfish, coral polyps, sea turtles, jellyfish, dolphins, dugongs, whales. **2** coral, fish, sharks, large fish, smaller fish, eels, crabs, prawns, sea stars, seahorses, clownfish, angelfish, coral polyps, sea turtles, jellyfish, dolphins, dugongs, whales. **3** seaweed, grasses, anemones. **4** Answers will vary (could be plankton, coral polyps and jellyfish). **5 a** plankton **b** dolphin **c** jellyfish **d** sea star **e** horse. **6** Answers will vary. **Check your understanding 1 b. 2 b. 3 a. 4 c. 5 b. 6 a. 7 a. 8 b.**

4 Plants

How do plants get food? **1** they may need more inorganic nutrients because the soil is poor quality. **2 a** minerals, absorb **b** glucose, carbon dioxide. **3 a** organic **b** inorganic **c** organic **d** organic **e** inorganic **f** inorganic **g** inorganic. **4 a** carbon dioxide, water **b** glucose. **5** Answers will vary but should include 'to produce energy' and 'to grow and survive'. **Inside plant cells 1** cell membrane, membrane, cell wall, cell wall, cytoplasm, nucleus, chloroplasts, chlorophyll, mitochondria, vacuoles. **2** Answers will vary. **Do plants breathe?** **1** carbon dioxide, humans, glucose, oxygen. **2** oxygen, oxygen, glucose, oxygen, carbon dioxide. **Photosynthesis 1** water, carbon dioxide. **2** glucose, oxygen. **3** glucose. **4** oxygen. **5** Sample answer: carbon dioxide + water + sunlight = glucose + oxygen. **6** green light, red light, blue light. **7 a** true **b** false **c** true **d** true **e** false **f** true. **Plant structures 1 a** bud **b** seeds **c** flower **d** leaves **e** stem **f** roots **g** xylem **h** phloem. **Check your understanding 1 b. 2 c. 3 b. 4 c. 5 a. 6 b. 7 b. 8 a.**

5 Animals

The importance of blood 1 a The network of vessels that carry an animal's blood is called the circulatory system. **b** Water, oxygen, nutrients and a way to remove waste products are four vital things that an animal's body needs to stay healthy. **c** Red blood cells

carry oxygen to the cells. **d** Plasma carries nutrients to the cells. **e** Plasma carries away waste products from the cells. **2 a** the cells of animals **b** stay healthy **c** have enough energy **d** oxygen is in the blood **e** the circulatory system. **3 a** water **b** oxygen **c** waste products **d** nutrients. **4 a** circulate **b** circulation **c** circulatory. **Turning food into energy 1** energy, digestive, bloodstream, absorption, glucose, humans. **3 a** cells **b** food **c** digestive **d** glucose **e** circulatory system **f** cells. **4 a** Animals and other living things need energy to survive. **b** The blood in the circulatory system carries glucose to all the body's cells. **c** Glucose plays an important part in the digestive system. **5** Answers will vary. **Waste systems 1** and takes away bad things from the cells, get their energy from glucose, and deliver waste products to other organs to be dealt with safely. **2 a** kidneys **b** skin **c** lungs. **Check your understanding 1 c. 2 b. 3 b. 4 a. 5 c. 6 a. 7 c. 8 c.**

6 Humans

Types of nutrients 1 a different **b** healthy **c** right **d** small **e** known. **2 a** digestive **b** circulatory **c** circulatory **d** digestive. **3 a** E **b** C **c** D **d** A. **4** natural, from, soil, fibre, cellulose, fruit, fats. **5** Answers will vary. **6 a** banana (it contains no protein) **b** potatoes (are starchy vegetables whereas all the others are leafy) **c** butter (is the only non-carbohydrate food in the list; it is a fat). **Body systems working together 1** the organ that pumps blood around the body, the lungs, cells that transmit electrical signals to and from the brain, oxygen. **2 a** heart, blood, cells **b** nerve cells **c** lungs, red. **3** lungs, nerve cell, bodies. **Removing waste products 1 a** waste products, water **b** waste water, sweat **c** excretory system. **2 a** carbon dioxide, breathes out, lungs **b** liver, kidneys, urine **c** sweat, skin, pores. **3 a** carbon dioxide **b** water **c** urea. **4 a** remove **b** assist **c** lungs **d** urine. **5 a** lungs, carbon dioxide **b** kidneys, urine **c** skin, sweat. **What else are bodies made of? 1** joints, skeletal, muscles, tendons, pears, stratens. **2** joints, skeletal, muscles, tendons, pairs, straightens. **Check your understanding 1 a. 2 b. 3 c. 4 a. 5 c. 6 b. 7 b. 8 b.**

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Topics include

Cells
Body systems and cells
Classifying living things

Plants
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Humans

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- Science understanding
 - Chemistry for years 7–8
 - Chemistry for years 9–10
 - Biology for years 7–8
 - Biology for years 9–10
- Science as a human endeavour
 - Biology for years 7–8
 - Biology for years 9–10
- Ecosystems
- Indigenous contexts
- Sustainability in the Asia Pacific

